

AthDGC - Athens Diachronic Glossa Chronos (Athens-PROIEL)

An open PROIEL-style treebank with
Indo-European parallels

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Background

Greek is the longest continuously attested member of the Indo-European family. The PROIEL Treebank (Haug & Jøhndal 2008; Eckhoff *et al.* 2018) established the gold standard for syntactically annotated parallel IE Bible corpora, but its scope is restricted to the New Testament. No openly licensed end-to-end pipeline currently covers the full diachronic record of Greek with explicit argument-structure tagging and

reproducible cross-lingual alignment.

Approach

AthDGC (“Athens-PROIEL”) is an open end-to-end pipeline that:

1. **Discovers** new Greek + parallel-language sources daily (archive.org, Perseus, First1K, Wikisource, Diorisis)

2. **Filters** via Greek-script ratio, apparatus-criticus rejection, content-hash dedup
 3. **Converts** to PROIEL XML 2.0
 4. **Annotates** with Stanza `grc_proiel` (+ `la_proiel`, `cu_proiel`, `got_proiel` for parallels)
 5. **Extracts** per-verb argument structure (subj, obj, oblique, voice, aspect)
 6. **Aligns** cross-lingually via LaBSE (sentence) + AwesomeAlign (word, mBERT)
 7. **Stores** as PROIEL XML + JSONL partitions + Qdrant vectors + Neo4j alignment graph
- 80+ TLG canonical authors with curated metadata

Argument-structure capture

Per VERB / AUX token AthDGC stores:

- subject, direct object, indirect / oblique args
- voice (active / middle / passive)
- aspect (perfective / imperfective)
- tense, mood, verb-form, person, number

stored in a column queryable across the cross-lingual alignment edges.

Coverage (v0.4)

- 89.9 M corpus rows
- 4.08 M annotated Greek rows
- 6,861 NT-aligned Greek verses
- 4 cross-aligned IE witnesses (lat, got, chu, xcl)
- 8 diachronic periods (Archaic to Modern)

Example query

“Show every Greek aorist transitive verb with an accusative object whose Latin Vulgate counterpart is a passive periphrastic”

Neo4j Cypher over Project: **CVL-CDSAML** at
 (:Token)-[:TRANSLATED_AS]->(NToken) Compute: GRNET ARIS
 edges, filtering source by (pa260305).
 (voice=Act,aspect=Perf,obj=Acc)
 and target by
 (voice=Pass,verb_form=Periphrastic).

Open access

- Source code (Apache-2.0):
github.com/AthDGC/Diachronic-Linguistics-Platform
- Dataset (CC-BY-4.0):
[10.5281/zenodo.20439182](https://doi.org/10.5281/zenodo.20439182)
- Showcase: athdgc.github.io
- HuggingFace:
[athdgc/diachronic-greek-corpus](https://huggingface.co/athdgc/diachronic-greek-corpus) (forthcoming)

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